

01 Variables and data types in Python

Introduction to Python – RSE course

WHY?

- python is language of modern tools used in data science, research, automation and ML
- Generative AI can **help** you code...
 - explaining error messages
 - suggesting fixes
 - generating code from scratch
 - even improving existing code

.... but AI is not always reliable

- Generates the **most likely text**, not guaranteed truth
- Can produce **plausible but incorrect code**
- Can make research results harder to **reproduce**

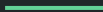
→ we have to understand the code

Jupyter notebook

easiest way to start with python
scripts

An interactive environment where
you can:

- write code
- run code in small blocks
- see the output
- combine code, text and
explanations



Variables

- sticky notes
- Use **variable = value** to assign a value to a variable in order to record it in memory
- Variables are created on demand whenever a value is assigned to them
- variable names
 - should be meaningful
 - letters, digits, underscores
 - are case-sensitive!



Calculations

- simple calculations

```
a = 3
b = 2
#addition
c = a + b
#subtraction
d = a - b
#multiplication
a = a * b
#division
f = a / b
#exponentiation
g = f ** b
```

- $g = ?$

Data types

- every object has a type
- common ones are:
 - integer numbers
 - floating point numbers,
 - strings, and
 - booleans.
- types control what operations can be done on objects

Data structures

- **List**

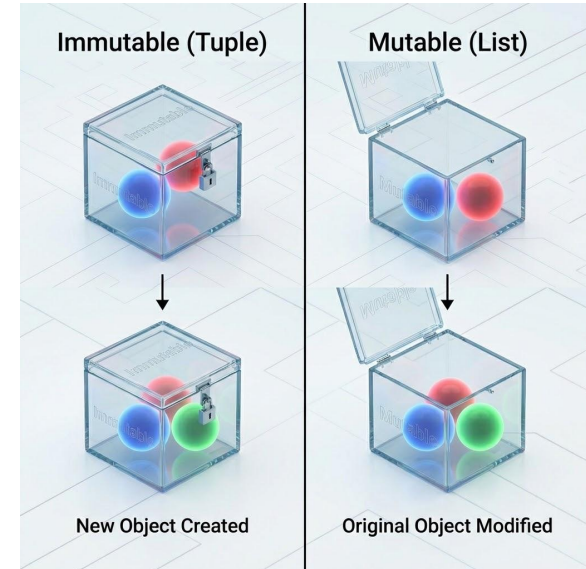
- stores many values in a single structure
- the values can be different data types
- the list indexing begins from 0
- `.append()`, `.remove()`, `.del()`

- **Tuple**

- fixed collection of value - coordinates

- **Dictionaries**

- stores key-value pairs
- structured data, records, JSON-like data



Key takeaways

- variables **store values**
 - every value has a **data type**
 - types control valid operations
 - **data structures** are used to store more values:
 - **list** - ordered, mutable
 - **tuple** - ordered, immutable
 - **dict** - key-value pairs
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